

Lifetime management of heart valve disease—treat it early and treat it right, first time



Heart valve disease is an important cause of death and disability, and its incidence is increasing due to the ageing population globally.¹ There is wide geographical variation in the cause of valvular disease, with rheumatic disease being a more prevalent cause in low-income and middle-income countries and degenerative diseases being a more prevalent cause in high-income countries. Presently, by the time most people are diagnosed and referred, the effect of treatment is often considered palliative rather than curative,² and survival without cardiac interventions for patients with severe symptoms is lower than for some malignant conditions (eg, prostate, bladder, and colorectal cancer).³ There is also wide global variation in potential access to treatment for heart valve disease, and there is global underprovision of high-quality treatment options.^{4,5} Recent decades have witnessed huge growth of novel transcatheter interventional treatments for structural heart disease, bringing the possibility of less invasive treatment options for patients rather than surgical approaches.^{1,6} Treating physicians have developed the concept of lifetime management, which suggests different combinations of surgical or interventional treatments.

Current evidence suggests that early referral for treatment of heart valve disease has the largest potential for improvement of patient survival to levels similar to the general population.^{7,8} However, the number of patients with heart valve disease (specifically mitral and tricuspid regurgitation) referred for invasive treatment is less than 25% of those who meet indications for referral, as seen both in English and American studies.^{4,5,9} This so-called referral inertia might be related to the perceived risks of treatment. Risk perception starts with patients, who often put symptoms of valvular disease (ie, shortness of breath and tiredness) down to signs of ageing and do not access help. The process continues with general physicians and conservative cardiologists, who might perceive the risks of intervention as higher than documented in trials. Finally, surgery generally evokes the greatest perception of risk among both patients and health-care providers, although short-term mortality often does not differ from that found for transcatheter interventions.¹⁰ If transcatheter interventions do show

lower short-term mortality compared with surgery, this difference disappears when patients are referred early.^{11,12} Yet, a limitation of low-risk and medium-risk trials is the lack of robust long-term data and the presence of biases (such as deviation from assigned treatment, performance of concomitant procedures, or relevance of endpoints) that apply to randomised trials.¹³ Furthermore, there are multiple real-life registries in Europe and the USA where long-term mortality is substantially (over 40%) lower in risk-adjusted patients with a surgical strategy,¹⁴ a dimension that is difficult to explain with biases alone. Despite these data, an increasing number of patients are being offered transcatheter procedures, as seen in the USA and Germany.¹⁵ This increase correlates with a gradual reduction in surgical procedure volumes, which carries the risk of suboptimal patient care.

Therefore, we recommend early referral of patients with heart valve disease to specialist heart networks that include a wide multidisciplinary team. This team can assess each patient to individualise and tailor treatment decisions that best suit lifetime management, balancing short-term and long-term outcomes and starting with the first intervention. Our suggested heart networks could be designed to be similar to tumour boards in oncology, where interventional cardiologists, surgeons, heart failure specialists, imaging specialists, and referring physicians can interact early to tailor the best treatment option for a patient. These networks could meet virtually and would facilitate early referral from generalists to specialists. Literature on aortic, mitral, and tricuspid valve procedures suggest that triaging the right patient to the right treatment at the right time can have prognostic implications for the individual patient and potential cost savings for health-care systems in the long term.^{7,16,17} With procedures like mitral valve or tricuspid valve repair,¹⁸ the Ross procedure,¹⁹ and newer mechanical valves that require lower levels of anticoagulation,²⁰ there are developments within cardiac surgery that are positively affecting long-term outcomes. We believe that heart networks (like tumour boards) need to assess both early and long-term outcomes. Our message for physicians involved in management of heart valve disease is to refer patients to heart teams early, and for heart teams to

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form wider heart networks to identify the most durable treatments at the lowest risk. Lifetime management for us means that when an invasive procedure is chosen, it is best done early and right, first time.

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